NAME : _____

STUDENT ID : _____ SECTION: _____

INSTRUCTIONS: Please answer all questions in the spaces given and show ALL your workings.

1. 2 bytes is equal to how many: [3m]
   a. Nibbles =
   b. Words =
   c. Bits =

2. Give ONE example of code in the digital systems. [1m]

3. Convert $\mathbf{90.3125_{10}}$ to

   a. Binary = [4m]

   b. Hexadecimal [2m]

4. Using 8-bits number system, change **-24** into its representation of:

   a. Sign and magnitude [2m]

   b. 1's complement [2m]

   c. 2's complement [1m]

5. Using 10-bits 2's complement signed numbers, perform the following operation.

   a. **24 + 16** [2m]

   b. **-90 - 16** [3m]